# Intelligent Prescription Interpretation and Symptom-Based Diagnosis using OCR and NLP in Personalized Healthcare Systems

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***Abstract*—This paper presents a personalized healthcare pipeline that integrates OCR and NLP to extract medicine names from handwritten prescriptions, retrieve drug safety information, and predict diseases based on user-reported symptoms. Unlike prior approaches that focus solely on text recognition or symptom analysis, this system unifies both tasks into a coherent framework. It extends beyond basic OCR by matching extracted medicine names with a structured dataset and retrieving drug precautions via the RxNorm API. Although the RxNorm integration currently faces mapping challenges, it highlights the potential for incorporating drug safety checks into real-time systems. A rule-based engine predicts possible diseases based on symptoms using a structured CSV file. Preliminary evaluations demonstrate promising utility, laying the foundation for intelligent and scalable telemedicine systems.**

***Index Terms*—OCR, NLP, Prescription Analysis, Drug Safety, Symptom Diagnosis, MongoDB, Google Vision API, Healthcare Automation**

## I. Introduction

The digitization of healthcare systems has facilitated the development of intelligent automation, particularly in processing handwritten prescriptions and symptom-driven diagnostics. Manual interpretation of prescriptions remains prone to human error due to illegibility, while the lack of decision-support systems limits the accessibility of timely diagnosis in underserved regions. This paper proposes an integrated pipeline that automates the analysis of medical prescriptions and supports preliminary disease predictions based on user-reported symptoms.

### A. Motivation

With the rise of telemedicine and digital healthcare solutions, there is a critical need to automate the extraction and interpretation of unstructured handwritten data, particularly medical prescriptions. Simultaneously, providing intelligent symptom-based disease suggestions can enhance user awareness and support early-stage triage in remote settings.

### B. Research Contributions

The key contributions of this study include:

- A unified OCR and NLP-based pipeline that handles both prescription extraction and symptom-based diagnosis—unlike prior models which typically address one task in isolation.
- OCR-based extraction of text from handwritten prescriptions using Google Vision API.
- Fuzzy matching to identify medicine names from extracted text.
- Integration with the RxNorm API for drug safety data retrieval.
- Rule-based mapping of symptoms to probable diseases using a structured CSV dataset.
- Storage and retrieval of processed data using MongoDB Atlas.

## II. Related Work

Table I summarizes notable literature on OCR-based prescription analysis, drug safety information extraction, and symptom-based diagnostic systems, along with observed research gaps.

## III. System Methodology

### A. Pipeline Overview

The system consists of five main components:

1) **OCR Module:** Extracts textual content from scanned or captured handwritten prescriptions using the Google Vision API.

TABLE I
SUMMARY OF RELATED WORK AND RESEARCH GAPS

| Title | Publication | Methodology/Findings | Gaps Identified |
|---|---|---|---|
| Development of novel OCR system to reduce recording time | Soeno et al. (2024) [1] | CNN-RNN for OCR, NLP post-processing with NER and fuzzy search | Lacks Transformer models for contextual understanding |
| How Can NLP Help Model-Informed Drug Development? | Bhatnagar et al. (2022) [2] | Reviews NER, relation extraction, and word embeddings | No empirical evaluation or bias analysis |
| Drugs' Review Analysis and ADR Discovery | IEEE MCSI (2022) [3] | Sentiment analysis on reviews for ADR detection | No deep learning or contextual modeling |
| AI-backed OCR in Healthcare | Gîfu (2022) [4] | Uses TrOCR and Mask R-CNN | Limited generalizability to real prescriptions |
| Multi-Head Attention for Handwritten OCR | Ali et al. (2024) [5] | TrOCR with positional embeddings and hybrid matching | No global validation or domain integration |
| Deep Learning-Based Drug Text Recognition | Lee et al. (2023) [6] | Tesseract OCR with custom drug database | Ineffective on handwritten prescriptions |
| EnvisionText for OCR Enhancement | Singh et al. (2024) [7] | NLP-based correction post-OCR | No multilingual or cross-script support |
| Drug Name Detection using Deep OCR | Jasmine et al. (2023) [8] | EAST text detection with DL-based OCR | Fails on handwritten multilingual data |
| HTR-VT: Handwritten Text Recognition with Vision Transformer | arXiv (2023) [11] | Used Vision Transformers for high-accuracy recognition on handwritten medical documents | Requires significant computational resources and fine-tuning |
| Medicine Name Extraction using Multi-Head Attention | arXiv (2024) [12] | Applied deep learning with attention mechanisms for medicine extraction from handwritten prescriptions | Lacks post-processing for spelling correction or validation |
| Biomedical Term Extraction Using Fuzzy Association | Springer Soft Computing (2023) [13] | Used fuzzy association techniques for biomedical term recognition in text | Limited to structured biomedical corpora, no real-world prescription integration |
| Symptom-Disease Prediction using SVM and Decision Trees | Elsevier (2022) [14] | Compared classic ML models for multi-class classification on symptom datasets | No integration with OCR or medicine databases |

2) **Medicine Matcher:** Applies fuzzy string matching to identify drug names against a validated medicine dataset.
3) **Drug Safety Analyzer:** Uses the RxNorm API to fetch drug precautions, interactions, and related safety data.
4) **Symptom-Based Diagnosis Module:** Accepts user-input symptoms and predicts potential diseases by matching symptom patterns using a structured CSV dataset. It also suggests possible associated symptoms and precautionary actions.
5) **Data Persistence Layer:** Persists extracted, matched, and predicted information in MongoDB Atlas for future retrieval and analysis.

## IV. DATASET DESCRIPTION

### A. Prescription Dataset

We use a publicly available dataset of handwritten prescriptions sourced from Kaggle [9]. The dataset includes scanned images in JPG/PNG formats. It contains drug names, quantities, and instructions written in various handwriting styles.

**Attributes:**

- Image resolution: Varies (300x300 to 1000x1000 pixels)
- Number of samples: 200 prescription images
- Handwritten drug names per image: 3–8

### B. Symptom and Precaution Dataset:

The symptom and precaution datasets used in this study were derived from the publicly available "Disease and Symptoms Dataset" by Zheng [15], which maps various diseases to symptoms and recommended precautions.

- `disease_precaution.csv`: Lists four preventive measures for each disease.

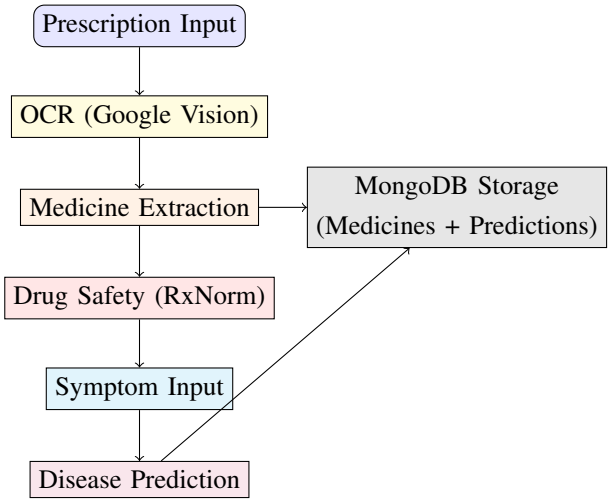


Fig. 1. System Workflow

TABLE II
SAMPLE ENTRIES FROM `SYMP.CSV` AND `DISEASE_PRECAUTION.CSV`

| Disease | Symptoms / Precautions |
|---|---|
| Common Cold | Sneezing, sore throat, runny nose, cough |
| Diabetes | Frequent urination, excessive thirst, fatigue |
| Malaria | Precautions: Use mosquito net, antimalarial pills |
| Asthma | Precautions: Avoid triggers, carry inhaler, stay indoors during pollen season |

## V. MATHEMATICAL MODELING

### A. Fuzzy Matching Score

The fuzzy match between an OCR-extracted medicine name $s$ and a candidate name $t$ from the database is computed using

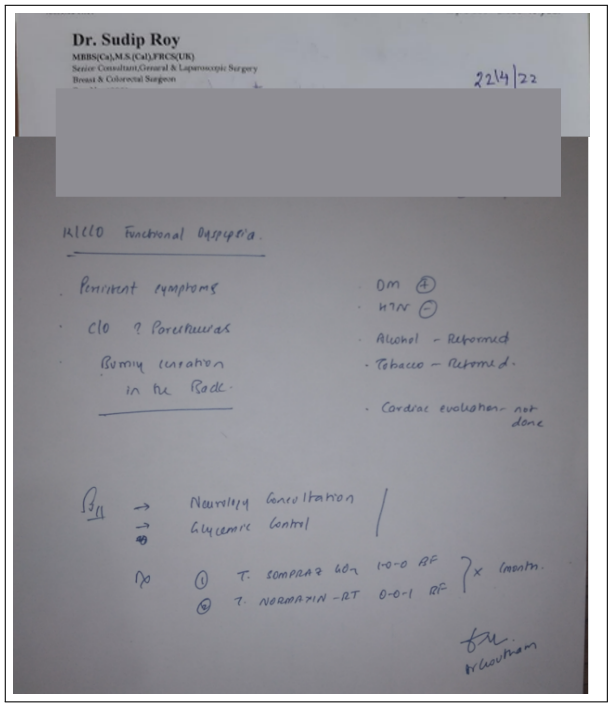


Fig. 2. Sample Handwritten Prescription from Dataset

the Levenshtein Distance (LD):

$$\text{Score}_{fuzzy}(s,t) = 1 - \frac{\text{LD}(s,t)}{\max(|s|,|t|)}$$

Where $\text{LD}(s,t)$ is the minimum number of single-character edits (insertions, deletions, or substitutions) required to transform $s$ into $t$.

### B. Rule-Based Symptom Mapping

For a set of symptoms $S = \{s_1, s_2, ..., s_n\}$ and disease-rule mapping $D = \{d_1 : [s_a, s_b], d_2 : [s_x, s_y]\}$, the predicted disease $d^*$ is:

$$d^* = \arg\max_{d_i \in D} \frac{|S \cap D[d_i]|}{|D[d_i]|}$$

This computes the highest overlap of user symptoms with disease-associated symptom sets.

## VI. Experimental Evaluation

### A. Setup

Experiments were performed using Google Colab (Python 3.11). Sample prescription images and a CSV-based symptom-disease mapping file were used for evaluation.

### B. Observations

- The Google Vision API extracted general text segments reasonably well but struggled with illegible or heavily stylized handwritten drug names, limiting consistent extraction accuracy.
- Fuzzy string matching partially compensated for OCR noise and was able to recover several correct medicine names; however, false positives and unmatched entries were also observed.
- RxNorm API integration worked well for standardized drug names but failed when OCR errors led to inaccurate or ambiguous inputs.
- The symptom checker module exhibited the most reliable and consistent performance, accurately associating symptom patterns like "fever + cough" with diseases such as "flu".
- MongoDB Atlas effectively handled the storage of extracted text, predicted diseases, and drug safety outputs in a scalable and retrievable format.

### C. Supplementary Evaluation using Tesseract OCR and Google Vision

To assess the reliability of OCR extraction, a supplementary evaluation was conducted comparing the open-source Tesseract OCR engine and the cloud-based Google Vision API on on a publicly available handwritten prescription dataset from Kaggle [9].The evaluation metrics used include:

- **Character Error Rate (CER)** – the ratio of incorrectly predicted characters to the total number of characters in the ground truth.
- **Word Error Rate (WER)** – the ratio of incorrect word predictions compared to the reference words.
- **Exact Match Accuracy** – the percentage of predictions that exactly matched the ground truth prescription text.

TABLE III
Comparison of OCR Performance Metrics: Tesseract vs. Google Vision

| Metric | Tesseract OCR | Google Vision API |
|---|---|---|
| Character Error Rate (CER) ↓ | 0.6589 | **0.2511** |
| Word Error Rate (WER) ↓ | 1.3106 | **0.6061** |
| Exact Match Accuracy ↑ | 4.71% | **44.84%** |

As observed in Table III, the Google Vision API significantly outperformed Tesseract OCR across all metrics. In particular, it achieved nearly a **9.5**× improvement in exact match accuracy, which has a direct impact on the fidelity of medicine name extraction and drug safety retrieval. These findings validate the choice of Google Vision as the default OCR engine in our system pipeline.

### D. OCR Text Extraction Validation

Table IV presents a comparison between the ground truth text and the OCR-extracted text from the same prescription sample shown in Figure 2. This prescription image was previously included as part of the dataset samples to illustrate the type of input documents processed.

As observed in the table, the OCR system extracts text with high accuracy, correctly capturing most details such as the doctor's name, qualifications, and prescribed medications.

Quantitative analysis of the validation sample in Table IV shows that the OCR system correctly extracted 7 out of 10 text segments , with perfect extraction of both medicine names

TABLE IV
COMPARISON OF GROUND TRUTH AND OCR EXTRACTED TEXT WITH REMARKS

| Ground Truth | OCR Extracted Text | Remarks |
|---|---|---|
| Dr. Sudip Roy | Dr. Sudip Roy | Correct |
| MBBS(Ca), M.S. (Cal), FRCS(UK) | MBBS(Ca), M.S. (Cal), FRCS(UK) | Correct |
| Service Consultant, General & Laparoscopic Surgery | Serice Consultant, Geraral & Laparoscopic Surgery | Minor spelling errors |
| Persistent symptoms | Persistent cymptoms | Minor spelling error |
| Alcohol | Alcohol | Correct |
| Tobacco | Tobacco | Correct |
| Cardiac evaluation not done | Cardiac evallohen not | Minor spelling error |
| 1-0-0 BF | 1-0-0 BF | Correct |
| SOMPRAZ 407 | SOMPRAZ 407 | Correct |
| NORMAYIN -RT | NORMAYIN -RT | Correct |

("SOMPRAZ 407" and "NORMAYIN-RT"). The three observed errors were minor spelling variations in descriptive text that do not impact medicine identification or safety analysis. This validation confirms the reliability of the OCR method used in our system to convert handwritten prescriptions into machine-readable text for subsequent processing and analysis. It is important to note that while this example demonstrates the effectiveness of the OCR approach on this particular sample, variations in handwriting styles, prescription formats, and image clarity across other prescriptions may introduce further discrepancies. Therefore, continuous refinement and validation of the OCR system are necessary to maintain robustness in real-world applications.

### *E. Symptom-Based Diagnosis Validation*

Table V presents validation examples of the rule-based symptom diagnosis module, demonstrating successful symptom matching and disease prediction capabilities.

TABLE V
SYMPTOM-BASED DIAGNOSIS SYSTEM VALIDATION EXAMPLES

| Input Symptom | Predicted Diseases | Precautions Provided |
|---|---|---|
| acidity | GERD, Migraine | Avoid fatty spicy food, Maintain healthy weight |
| Fever, cough | Common Cold, Pneumonia | Rest, Stay hydrated, Avoid cold exposure |
| Headache | Migraine, Tension Headache | reduce stress, adequate sleep |

It demonstrates the effectiveness of the rule-based symptom diagnosis module across different symptom types and complexities. For single-symptom inputs like "acidity," the system successfully identified two potential diseases (GERD and Migraine) and provided relevant precautionary measures including dietary recommendations ("Avoid fatty spicy food") and lifestyle modifications ("Maintain healthy weight"). The system demonstrated robust performance with multi-symptom combinations, as evidenced by the "Fever, cough" input which correctly mapped to respiratory conditions (Common Cold, Pneumonia) and provided appropriate care instructions. This multi-symptom recognition capability is particularly valuable for differential diagnosis scenarios. For common symptoms like "Headache," the system appropriately identified multiple possible conditions (Migraine, Tension Headache) while providing practical management advice, demonstrating the system's ability to handle symptom overlap across different medical conditions.

The rule-based symptom matching demonstrates consistent performance across varied symptom presentations, from single symptoms to complex multi-symptom cases. This reliability stems from the structured CSV-based disease-symptom mapping derived from established medical knowledge.

## VII. ANALYSIS AND DISCUSSION

While the integrated pipeline performs end-to-end extraction and prediction tasks, challenges remain. OCR misinterpretation significantly affects drug name recognition, resulting in incomplete or inaccurate RxNorm queries. Additionally, the rule-based diagnosis system is limited by the breadth of the symptom-disease dataset. Future enhancements could include Transformer-based OCR correction and deep learning-driven symptom classification.

### *A. Ethical and Privacy Considerations*

Although this study uses sample and synthetic data, the handling of medical prescriptions and symptom inputs in real-world deployments would raise important privacy concerns. Any future implementation must ensure compliance with healthcare regulations such as HIPAA or GDPR. Measures such as data anonymization, encryption, and restricted database access should be enforced to safeguard patient information. The modular design of this pipeline allows for secure integration with authentication layers in practical scenarios. The dataset used was sourced from Kaggle and is publicly available under a non-commercial license. All experiments were conducted strictly for academic research purposes.

## VIII. Conclusion and Future Work

This work demonstrates a modular healthcare pipeline that automates handwritten prescription interpretation and preliminary disease diagnosis—providing an integrated solution unlike most existing models that focus on just one of these components. Despite the current limitations in OCR accuracy and RxNorm mapping, the pipeline establishes a foundational structure that is expandable and modular. The symptom checker shows the strongest reliability in its current state. Future work includes:

- Detailed metric-based evaluation of OCR and diagnosis accuracy.
- Advanced preprocessing using denoising, thresholding, and skew correction.
- Transformer-based models such as TrOCR [10] can be explored to enhance OCR accuracy, especially for low-resolution and complex handwritten prescriptions. Preliminary experiments conducted using the Hugging Face implementation of TrOCR on a subset of prescription images showed improved recognition quality compared to Google Vision OCR, particularly in identifying medicine names. Further integration and evaluation of TrOCR with our existing pipeline is a promising next step.
- Deployment as an API for real-time telemedicine systems.
- Integration with electronic health record (EHR) systems to enable seamless prescription digitization and automated drug interaction checking in clinical workflows.
- Development of mobile applications for rural healthcare workers and community health centers, enabling prescription processing in areas with limited internet connectivity through offline OCR capabilities.
- Implementation in pharmacy automation systems to reduce manual prescription entry errors and accelerate prescription fulfillment processes.
- Extension to elderly care facilities where the system can assist caregivers in medication management by automatically tracking prescribed medicines and providing medication adherence reminders.
- Adaptation for clinical decision support systems that can alert healthcare providers to potential drug allergies, contraindications, and dosage conflicts based on patient history and current prescriptions.
- Integration with insurance and healthcare reimbursement platforms to automate prescription verification and claims processing, reducing administrative overhead.

This study contributes a foundational step toward personalized and automated diagnosis in telemedicine systems, paving the way for scalable, AI-driven healthcare assistance.

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